

2.1. Explain why the size of the hypothesis space in the *EnjoySport* learning task is 973. How would the number of possible instances and possible hypotheses increase with the addition of the attribute *WaterCurrent*, which can take on the values *Light*, *Moderate*, or *Strong*? More generally, how does the number of possible instances and hypotheses grow with the addition of a new attribute *A* that takes on *k* possible values?

Answer:

1. Because we only consider semantically distinct hypotheses. $973 = 1 + 4 * 3 * 3 * 3 * 3 * 3$, 1 for $\{\emptyset\}$, and $4 * 3 * 3 * 3 * 3 * 3$ for combinations of every feature's value which including '?'
2. If we add *WaterCurrent* then hypotheses will grow to $1 + 4 * 3 * 3 * 3 * 3 * 3 * 4 = 3889$, increased by $3889 - 973 = 2916$. And instances will grow to $3 * 2 * 2 * 2 * 2 * 2 * 3 = 288$, increased by $288 - 96 = 192$.
3. For feature has *k* value, the hypotheses will increase by $(\text{origin} - 1) * k$, and instances will increase by $\text{origin} * (k - 1)$

2.2. Give the sequence of S and G boundary sets computed by the CANDIDATE-ELIMINATION algorithm if it is given the sequence of training examples from Table 2.1 in reverse order. Although the final version space will be the same regardless of the sequence of examples (why?), the sets S and G computed at intermediate stages will, of course, depend on this sequence. Can you come up with ideas for ordering the training examples to minimize the sum of the sizes of these intermediate S and G sets for the H used in the *EnjoySport* example?

Answer:

Initial:

- $S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \rangle$.
- $G_0 = \langle ?, ?, ?, ?, ? \rangle$.

Example1: <sunny, warm, high, strong, cool, change, yes>

- $S_1 = \langle \text{sunny, warm, high, strong, cool, change} \rangle$
- $G_1 = \langle ?, ?, ?, ?, ? \rangle$

Example2: <rainy, cold, high, strong, warm, change, no>

- $S_2 = \langle \text{sunny, warm, high, strong, cool, change} \rangle$
- $G_2 = \langle \text{sunny, ?, ?, ?, ?}, \langle ?, \text{warm, ?, ?, ?} \rangle, \langle ?, ?, ?, \text{cool, ?} \rangle$

Example3: <sunny, warm, normal, strong, warm, same, yes>

- $S_3 = \langle \text{sunny, warm, ?, strong, ?, ?} \rangle$

- $G3 = \langle \text{sunny}, ?, ?, ?, ? \rangle, \langle ?, \text{warm}, ?, ?, ? \rangle$

Example4: $\langle \text{sunny}, \text{warm}, \text{high}, \text{strong}, \text{warm}, \text{same}, \text{yes} \rangle$

- $S4 = \langle \text{sunny}, \text{warm}, ?, \text{strong}, ?, ? \rangle$
- $G4 = \langle \text{sunny}, ?, ?, ?, ? \rangle, \langle ?, \text{warm}, ?, ?, ? \rangle$

Why the final version space will be the same regardless of the sequence of examples?

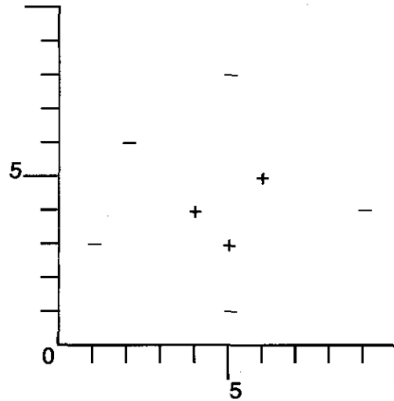
- Because the final S is the most special hypotheses for all samples, and final G is the least General hypotheses for all samples. So it doesn't care about the sequence.

Can you come up with ideas for ordering the training examples to minimize the sum of the sizes of these intermediate S and G sets for the H used in the *EnjoySport* example?

- **Train positive example first**, because we find that the reason of increasing size is the negative example generate too many G, but G is restrained by S(G must be general than S).

2.4. Consider the instance space consisting of integer points in the x, y plane and the set of hypotheses H consisting of rectangles. More precisely, hypotheses are of the form $a \leq x \leq b, c \leq y \leq d$, where a, b, c , and d can be any integers.

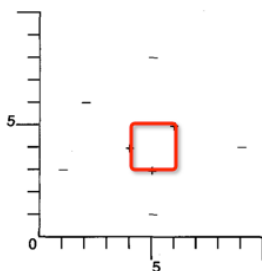
- (a) Consider the version space with respect to the set of positive (+) and negative (-) training examples shown below. What is the S boundary of the version space in this case? Write out the hypotheses and draw them in on the diagram.



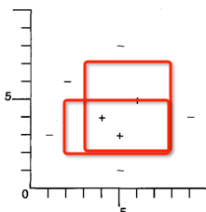
- (b) What is the G boundary of this version space? Write out the hypotheses and draw them in.
- (c) Suppose the learner may now suggest a new x, y instance and ask the trainer for its classification. Suggest a query guaranteed to reduce the size of the version space, regardless of how the trainer classifies it. Suggest one that will not.
- (d) Now assume you are a teacher, attempting to teach a particular target concept (e.g., $3 \leq x \leq 5, 2 \leq y \leq 9$). What is the smallest number of training examples you can provide so that the CANDIDATE-ELIMINATION algorithm will perfectly learn the target concept?

Answer:

- a) $S = \{4 \leq x \leq 6, 3 \leq y \leq 5\}$



- b) $G = \{3 \leq x \leq 8, 2 \leq y \leq 7\}, \{2 \leq x \leq 8, 2 \leq y \leq 5\}$



- c) 1. must have a shrinking: (3, 5)
2. may not have a shrinking: (5, 4)

d) 4 points: (3,2, +) (5, 9, +) and (2, 1, -), (6, 10, -)

2.5. Consider the following sequence of positive and negative training examples describing the concept “pairs of people who live in the same house.” Each training example describes an *ordered* pair of people, with each person described by their sex, hair

color (black, brown, or blonde), height (tall, medium, or short), and nationality (US, French, German, Irish, Indian, Japanese, or Portuguese).

- + $\langle\langle \text{male brown tall US} \rangle\langle \text{female black short US} \rangle\rangle$
- + $\langle\langle \text{male brown short French} \rangle\langle \text{female black short US} \rangle\rangle$
- $\langle\langle \text{female brown tall German} \rangle\langle \text{female black short Indian} \rangle\rangle$
- + $\langle\langle \text{male brown tall Irish} \rangle\langle \text{female brown short Irish} \rangle\rangle$

Consider a hypothesis space defined over these instances, in which each hypothesis is represented by a pair of 4-tuples, and where each attribute constraint may be a specific value, “?”, or “ $\emptyset$,” just as in the *EnjoySport* hypothesis representation. For example, the hypothesis

$\langle\langle \text{male ? tall ?} \rangle\langle \text{female ? ? Japanese} \rangle\rangle$

represents the set of all pairs of people where the first is a tall male (of any nationality and hair color), and the second is a Japanese female (of any hair color and height).

- (a) Provide a hand trace of the CANDIDATE-ELIMINATION algorithm learning from the above training examples and hypothesis language. In particular, show the specific and general boundaries of the version space after it has processed the first training example, then the second training example, etc.
- (b) How many distinct hypotheses from the given hypothesis space are consistent with the following single positive training example?

+ $\langle\langle \text{male black short Portuguese} \rangle\langle \text{female blonde tall Indian} \rangle\rangle$

- (c) Assume the learner has encountered only the positive example from part (b), and that it is now allowed to query the trainer by generating any instance and asking the trainer to classify it. Give a specific sequence of queries that assures the learner will converge to the single correct hypothesis, whatever it may be (assuming that the target concept is describable within the given hypothesis language). Give the shortest sequence of queries you can find. How does the length of this sequence relate to your answer to question (b)?
- (d) Note that this hypothesis language cannot express all concepts that can be defined over the instances (i.e., we can define sets of positive and negative examples for which there is no corresponding describable hypothesis). If we were to enrich the language so that it *could* express all concepts that can be defined over the instance language, then how would your answer to (c) change?

Answer:

a)

Initial:

- $S_0 = \langle \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle, \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle \rangle$.
- $G_0 = \langle \langle ?, ?, ?, ? \rangle, \langle ?, ?, ?, ? \rangle \rangle$.

Example1: + $\langle \langle \text{male, brown, tall, US} \rangle, \langle \text{female, black, short, US} \rangle \rangle$

- $S_1 = \langle \langle \text{male, brown, tall, US} \rangle, \langle \text{female, black, short, US} \rangle \rangle$
- $G_1 = \langle \langle ?, ?, ?, ? \rangle, \langle ?, ?, ?, ? \rangle \rangle$.

Example2: + $\langle \langle \text{male, brown, short, French} \rangle, \langle \text{female, black, short, US} \rangle \rangle$

- $S_2 = \langle \langle \text{male, brown, ?, ?} \rangle, \langle \text{female, black, short, US} \rangle \rangle$
- $G_2 = \langle \langle ?, ?, ?, ? \rangle, \langle ?, ?, ?, ? \rangle \rangle$.

Example3: + $\langle \langle \text{male, brown, tall, Irish} \rangle, \langle \text{female, brown, short, Irish} \rangle \rangle$

- $S_3 = \langle \langle \text{male, brown, ?, ?} \rangle, \langle \text{female, ?, short, ?} \rangle \rangle$
- $G_3 = \langle \langle ?, ?, ?, ? \rangle, \langle ?, ?, ?, ? \rangle \rangle$.

Example4: - $\langle \langle \text{female, brown, tall, German} \rangle, \langle \text{female, black, short, Indian} \rangle \rangle$

- $S_4 = \langle \langle \text{male, brown, ?, ?} \rangle, \langle \text{female, ?, short, ?} \rangle \rangle$
- $G_4 = \langle \langle \text{male, ?, ?, ?} \rangle, \langle ?, ?, ?, ? \rangle \rangle$.

b)

- According to Version Space in answer question a)
- $\langle \langle \text{male, ?, ?, ?} \rangle, \langle \text{female, ?, ?, ?} \rangle \rangle$
- $\langle \langle \text{male, ?, ?, ?} \rangle, \langle ?, ?, ?, ? \rangle \rangle$
- There are only **2** hypotheses can accept this example.

c)

- We can try as many as different value of all feature at one time. This means this example need to satisfy G and be most different to all given example. Because in this hypotheses space, if one feature can classify two different value to positive sample, then we can set the hypotheses of this feature to '?'. So if we get:

- $\langle \langle \text{male, black, short, Portuguese} \rangle, \langle \text{female, blonde, tall, Indian} \rangle \rangle +$
- $\langle \langle \text{female, brown, tall, US} \rangle, \langle \text{male, black, short, US} \rangle \rangle +$

Then we can get final hypotheses as $\langle \langle ?, ?, ?, ? \rangle, \langle ?, ?, ?, ? \rangle \rangle$. But this is not for sure we can get correct hypotheses. We need try until we can't find a hypotheses have different value with all existed samples and satisfying G.

d)

If we design a hypotheses space that contains all probable hypotheses, we get an unbiased learner. Because of the futility of Bias-Free learning, we get a new hypotheses every time we give a new example.

2.6. Complete the proof of the version space representation theorem (Theorem 2.1).

- **Version space representation theorem.** Let X be an arbitrary set of instances and let H be a set of Boolean-valued hypotheses defined over X . Let $c : X \rightarrow \{0, 1\}$ be an arbitrary target concept defined over X , and let D be an arbitrary set of training examples $\{ \langle x, c(x) \rangle \}$. For all X, H, c , and D such that S and G are well defined.

$$\blacksquare \quad VS_{H,D} = \{h \in H \mid (\exists s \in S) (\exists g \in G) (g \succeq_g h \succeq_g s)\}$$

Answer:

Proof.

To prove the theorem it suffices to show that (1) every h satisfying the right-hand side of the above expression is in $VS_{H,D}$ and (2) every member of $VS_{H,D}$ satisfies the right-hand side of the expression.

1. Let g be an arbitrary member of G , s be an arbitrary member of S , and h be an arbitrary member of H , such that $g \succeq_g h \succeq_g s$. Then by the definition of S , s must be satisfied by all positive examples in D . Because $h \succeq_g s$, h must also be satisfied by all positive examples in D . Similarly, by the definition of G , g cannot be satisfied by any negative example in D . Because h is satisfied by all positive examples in D and by no negative examples in D , h is consistent with D , and therefore h is a member of $VS_{H,D}$.
2. Assume h is a hypotheses in $VS_{H,D}$ and h is not in the right-hand side. According to the definition of $VS_{H,D}$, h must be satisfied by all positive and negative examples. But h is not in the right hand side, so $h <_g s$ or $h > g$. If $h <_g s$, then $_g s$ must not be the most specific hypotheses. If $h > g$, then g must not be the most general hypothesis. So there comes inconsistency.